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Original Article

Impact of Early Management in Osteoarticular Infections in Infants and Neonates: A Two-Year Retrospective Study

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Abstract

Background: Osteoarticular infections in neonates and infants, including osteomyelitis and septic arthritis, can cause irreversible joint damage, growth disturbance, deformity, and functional disability. Early recognition and definitive institutional treatment are therefore important. This study evaluated the clinical characteristics, risk factors, causative organisms, treatment, and relationship between delay in institutional presentation and outcome. **Methods:** This retrospective study included 40 eligible neonates and infants treated between June 2019 and May 2021. Data were obtained from the institutional Hosp Gestor database after Institutional Review Board approval and informed consent. Delay in institutional presentation was categorized as 0–7 days, 8–30 days, 1–3 months, and >3 months. Demographic characteristics, infection type and site, risk factors, laboratory findings, microbiology, treatment, and sequelae were analyzed. Statistical analysis was performed using SPSS 27.0 and GraphPad Prism 5; chi-square or Fisher's exact test was planned for categorical comparisons, with $P \leq 0.05$ considered statistically significant. **Results:** Of 40 patients, 8 (20%) were neonates (<28 days). Males constituted 70% of the cohort and the mean age was 5.4 months. Septic arthritis was the most common diagnosis (53%), followed by osteomyelitis (43%) and concomitant septic arthritis with osteomyelitis (4%). The hip was the most frequent site of septic arthritis (76.2%). Seventeen patients (42.5%) had a history of previous infection or trauma; pulmonary disease accounted for 9 of these (52.9%). Atypical associated factors included preterm delivery/low birth weight, congenital talipes equinovarus, intramuscular injection, scabies, and bacterial conjunctivitis. Eleven patients (27.5%) had received treatment elsewhere, though details of agents and duration were unavailable. *Staphylococcus aureus* was the commonest isolated organism (35%); *Acinetobacter haemolyticus* was isolated in 2.5%. Surgical cleaning/arthrotomy and drainage was the commonest procedure. Institutional delay was categorized as 0–7 days (19 patients), 8–30 days (10), 1–3 months (8), and >3 months (3). The most severe documented sequelae occurred with prolonged delay, including epiphyseal growth arrest, hip dysplasia, chronic osteomyelitis with recurrent pus formation, and gap non-union; category-wise complication counts were not available in the available dataset, so a valid inferential p-value for delay versus outcome could not be calculated. **Conclusion:** The most severe documented sequelae occurred among patients with prolonged delay in institutional presentation. Because category-wise complication counts were unavailable, the observed pattern should be interpreted descriptively rather than as a statistically confirmed association. The findings support prompt recognition and referral of neonatal and infant osteoarticular infections, particularly in settings where delayed presentation and treatment by informal health practitioners may occur.

Key Words: Infants, Neonates, Osteoarticular infection, *Staphylococcus aureus*, Time-to-treatment, Treatment outcome

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I. Introduction

Osteoarticular infections in infants and neonates, including osteomyelitis and septic arthritis, may progress rapidly and can result in systemic illness, irreversible joint damage, growth disturbance, limb deformity, and functional disability. Early treatment is therefore important to reduce permanent damage and sequelae[1-3].

In clinical practice, delayed institutional presentation may occur because of limited awareness, initial treatment outside formal healthcare facilities, referral delays, or failure to recognize early manifestations. Early radiographs may also be normal, making clinical assessment and appropriate use of ultrasonography or magnetic resonance imaging important when septic arthritis or osteomyelitis is suspected.

The present study evaluates the demographic characteristics, atypical presentations, risk factors, causative organisms, treatment patterns, and outcomes of osteoarticular infections in neonates and infants, with particular attention to the timing of institutional presentation.

Objectives

1. To identify demographic characteristics, atypical presentations, risk factors, and causative organisms in neonatal and infant osteoarticular infections.
2. To assess the role of definitive institutional management and document associated sequelae.
3. To compare clinical outcomes across categories of delay in institutional presentation.

II. Materials and Methods

Study design and setting

This was a retrospective observational study conducted after Institutional Review Board approval. Patients were identified from the institutional Hosp Gestor database. The study period was June 2019 to May 2021.

Participants

Forty eligible patients were included. Six additional patients were excluded because of incomplete charts. The 40 included patients were not a consecutive institutional series, as case selection depended on the available eligible records. Neonates were defined as patients younger than 28 days.

Inclusion and exclusion criteria

Neonates and infants admitted during the study period with a clinical and laboratory diagnosis of osteoarticular infection were eligible. Patients with incomplete charts were excluded.

Variables and definitions

Variables included age, sex, type and anatomical location of infection, delay in institutional presentation, previous infection or trauma, comorbidities and other reported risk factors, laboratory findings, microbiological results, treatment, complications, and sequelae. Delay in institutional presentation was categorized as 0–7 days, 8–30 days, 1–3 months, and >3 months.

Sample size

The sample size was calculated using Epi Info version 3.5.3, resulting in a target sample size

of 40 patients.

Statistical analysis

Data were entered into Microsoft Excel and analyzed using SPSS version 27.0 and GraphPad Prism version 5. Continuous variables were summarized using mean and standard deviation; categorical variables were summarized using counts and percentages. Chi-square or Fisher's exact test was planned for comparison of categorical variables, as appropriate, with $P \leq 0.05$ considered statistically significant. Because the number of complicated and uncomplicated patients within each delay category was not available, an inferential comparison of delay categories with complication status could not be performed.

III. Results

The final study sample comprised 40 patients. Eight (20%) were neonates younger than 28 days. Males predominated (70%), and the mean age was 5.4 months.

Diagnosis	n	%
Septic arthritis	21	52.5
Osteomyelitis	17	42.5
Septic arthritis + osteomyelitis	2	5.0
Total	40	100

Table 1: *Types of osteoarticular infection (n=40)*

Septic arthritis was the most common diagnosis. The hip was the most frequent site of septic arthritis, involving 76.2% of affected patients. Osteomyelitis most commonly involved the tibia.

Risk factors and previous infections

Seventeen patients (42.5%) had a history of previous infection or trauma. Pulmonary diseases, including bronchiolitis and pneumonia, were the most frequent previous infections, accounting for 57.14% (n=9) of these histories. Atypical associated factors included preterm delivery/low birth weight (20%), congenital talipes equinovarus (5%), previous intramuscular injection (5%), scabies (5%), and bacterial conjunctivitis (5%).

Treatment before institutional presentation

Twenty-nine patients (72.5%) had not received treatment before their first presentation. Eleven patients (27.5%) had received some form of treatment outside the institution before presentation. The specific antimicrobial agents, route, and duration of treatment were not available in the records supplied for analysis; the effect of prior antimicrobial exposure on culture yield could therefore not be reliably assessed.

Microbiology and laboratory findings

Staphylococcus aureus was the most commonly isolated organism (35%). *Acinetobacter haemolyticus* was identified as an atypical organism in 2.5% of cases. CRP and ESR were reported to be elevated in the study population.

Delay in institutional presentation and outcome

The available clinical records demonstrated progressively more severe sequelae with prolonged delay. Documented complications included epiphyseal growth arrest, dysplasia of the hip, chronic osteomyelitis with recurrent pus formation, and gap non-union. The most severe outcomes were observed among patients presenting after prolonged delay, particularly those with delay exceeding 3 months. Because the supplied data do not specify the number of complicated versus uncomplicated patients within each delay category, an exact chi-square/Fisher's exact p-value cannot be reported without introducing unsupported assumptions.

Delay category	n	%
0–7 days	19	47.5
8–30 days	10	25.0
1–3 months	8	20.0
>3 months	3	7.5
Total	40	100

Table 2: *Delay in institutional presentation (n=40)*

Treatment

Method	Description
Surgical cleaning/incision & drainage	Commonest operative approach
Arthrotomy and drainage	Common, particularly for septic arthritis
Intravenous antibiotics	Used as part of institutional treatment
Saucerization/sequestrectomy	Used where indicated
Debridement	Used where indicated
Osteotomy	Used for selected residual deformity

Table 3: *Treatment methods reported in the study*

IV. Discussion

The present study highlights the clinical importance of timely institutional management of osteoarticular infections in neonates and infants. Septic arthritis was the predominant infection, and the hip was the most frequently involved joint. The predominance of *Staphylococcus aureus* is consistent with the established microbiology of paediatric osteoarticular infection[4-9].

The study also identified potentially relevant preceding conditions, including pulmonary infections, preterm delivery and low birth weight, congenital talipes equinovarus, intramuscular injection, scabies, and bacterial conjunctivitis. These findings may help clinicians maintain a high index of suspicion in infants with fever, irritability, limb non-use, swelling, or reduced joint movement.

Delayed presentation in this study population was associated descriptively with more severe sequelae. Documented reasons for delay included lack of awareness and inappropriate treatment by informal health practitioners, which may contribute to delayed referral and prolonged infection before definitive treatment. In settings with limited access to specialist care, early recognition and referral are therefore particularly important.

Kocher et al. described clinical approaches to diagnosis and management of septic arthritis of the hip, while other studies have emphasized early antimicrobial therapy and appropriate surgical drainage when indicated[7-9]. Empirical antimicrobial therapy is generally initiated after appropriate cultures have been obtained when clinically feasible, with subsequent adjustment according to microbiological results and clinical response.

Early radiographs may not demonstrate osteoarticular infection, particularly in the initial stages. Ultrasonography is useful for detecting joint effusion and guiding aspiration in suspected septic arthritis, especially at the hip. Magnetic resonance imaging is more sensitive for early marrow, metaphyseal, epiphyseal, and soft-tissue involvement and can be considered when the diagnosis remains uncertain or osteomyelitis is suspected despite non-diagnostic initial radiographs.

The study's observation of severe sequelae after prolonged delay is clinically important, because damage to the growing epiphysis and physis may produce growth arrest, deformity, dysplasia, limb-length discrepancy, and persistent functional impairment even after eradication of infection.

Limitations

This study has several limitations. The sample size was small (40 patients), the study was conducted at a single institution, and the included cases were not a consecutive series, introducing the possibility of selection bias. The study was retrospective and depended on information available in the Hosp Gestor database; six patients with incomplete charts were excluded. Details of treatment received outside the institution, including antibiotic type and duration, were unavailable, limiting assessment of its effect on culture yield and outcomes. Reasons for delayed presentation were not systematically quantified. Long-term follow-up beyond the documented sequelae was limited. Finally, the supplied aggregate data did not provide category-wise numbers of complicated and uncomplicated patients, preventing calculation of a valid inferential p-value for the association between delay category and outcome.

V. Conclusion

Osteoarticular infections in neonates and infants can result in serious and irreversible sequelae when recognition and definitive treatment are delayed. This study demonstrates that prolonged delay in institutional presentation was accompanied by more severe documented complications, including epiphyseal growth arrest, hip dysplasia, chronic osteomyelitis with recurrent pus formation, and gap non-union. Lack of awareness and inappropriate treatment by informal health practitioners were identified as important contributors to delay. Early clinical suspicion, appropriate imaging and microbiological evaluation, prompt referral, and timely definitive institutional management are essential to improve outcomes.

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Case Study / Case Report

Emerging Clinical Significance of *Ochrobactrum anthropi* Septicemia in a Multimorbid ICU Patient: A Clinical Case Report

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Abstract

Ochrobactrum anthropi is a rare opportunistic Gram-negative, non-fermenting bacillus increasingly recognized as a cause of nosocomial infections, particularly in critically ill patients with indwelling catheters. Its ability to form biofilms and adhere to synthetic materials makes catheter-related bloodstream infections difficult to treat. We report a case of persistent *O. anthropi* bacteraemia in a hospitalized patient with a central venous catheter who presented with recurrent fever. Blood culture identified *O. anthropi* with susceptibility to carbapenems and resistance to several commonly used antibiotics. The patient was initially treated with piperacillin-tazobactam and later escalated to meropenem based on culture sensitivity. Despite partial clinical improvement, repeat blood cultures remained positive. Removal of the central venous catheter resulted in sterile repeat cultures, complete resolution of fever, and significant clinical recovery, highlighting the importance of prompt source control.

Key Words: Antimicrobial resistance, Biofilm formation, Central venous catheter, *Ochrobactrum anthropi* bacteraemia, Opportunistic nosocomial infection, Source control

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I. Introduction

Ochrobactrum anthropi is a Gram-negative, non-fermenting bacillus increasingly recognized as an opportunistic cause of healthcare-associated bloodstream infections. Although generally considered a low-virulence organism, it can cause clinically significant infection in critically ill patients and those with indwelling intravascular devices. Its ability to persist on foreign surfaces and its variable antimicrobial susceptibility may contribute to persistent bacteraemia

and therapeutic challenges. We report a case of persistent *O. anthropi* bacteraemia in a multimorbid critically ill patient with a central venous catheter, in whom bacteraemia persisted despite antimicrobial therapy and resolved following catheter removal. The case highlights the importance of recognizing an indwelling catheter as a potential source and the role of prompt source control in management.

II. Case Report

A sixty-three-year-old male with a central venous catheter in situ along the right internal jugular vein presented to the emergency department with shortness of breath (mMRC grade 4) and confusion for 2 days. He had a significant medical history of COPD, hypertension, type 2 diabetes mellitus, and chronic kidney disease, and a surgical history of permanent pacemaker implantation in 2017. He had previously been admitted for dyselectrolytaemia 15 days earlier. On general physical examination, the patient was disoriented and febrile (temperature 101°F). Vitals were BP 156/90 mmHg, PR 90 bpm, SpO_2 86%. He had bilateral pedal oedema, warm and moist extremities, and bilateral crepitations on auscultation. The patient was managed in the ICU. Blood samples for culture were collected before antibiotics were administered, and intravenous piperacillin-tazobactam was started.

Haematological investigation revealed haemoglobin 8.9 g/dL, leukocytosis (total leukocyte count 19,900/mm^3, 93% neutrophils, with toxic granules), and platelet count 95,000/mm^3. ESR was 36 mm/hour and CRP was positive. Renal function tests showed sodium 134 mmol/L, potassium 4.7 mmol/L, creatinine 3.3 mg/dL, urea 89 mg/dL, and uric acid 10.8 mg/dL. Urinalysis showed albuminuria with 20–21 pus cells/hpf. Chest X-ray (PA view) showed bilateral patchy opacities, and echocardiography showed severe pulmonary hypertension.

Blood cultures, processed on the automated BacT/ALERT system, identified *Ochrobactrum anthropi*. Empirical piperacillin-tazobactam (3.375 g IV every 6 h, extended infusion) was escalated to meropenem (1 g IV every 12 h) for seven days once sensitivity confirmed susceptibility to carbapenems. Although the patient showed clinical improvement with reduced symptoms and fewer febrile episodes, intermittent fever persisted. A repeat blood culture again showed persistent growth of *O. anthropi*, indicating ongoing catheter-related bacteraemia despite appropriate antimicrobial therapy. Given the organism's known biofilm-forming ability and persistent device colonization, the central venous catheter was removed. Following removal, repeat blood cultures became sterile, the bacteraemia resolved completely, and the patient became afebrile with marked clinical improvement.

III. Discussion

Ochrobactrum anthropi infection (reclassified in some literature as *Brucella anthropi*) is a Gram-negative, motile, non-fastidious, non-fermenting bacillus with strict oxidative metabolism belonging to the family Brucellaceae[1,2]. Although considered a low-virulence organism, its clinical significance lies in its ability to form biofilms and survive in antiseptic solutions, making it an important opportunistic nosocomial pathogen.

The organism is commonly isolated from hospital water sources and contaminated medical environments, including normal saline, antiseptic solutions, dialysis fluids, and indwelling medical devices such as central venous catheters and drainage tubes. Its strong adherence to foreign bodies and synthetic surfaces contributes significantly to persistent hospital-acquired infections. Beyond bacteraemia, other reported manifestations include infective endocarditis, meningitis, peritonitis, septic arthritis, and soft tissue infections[3].

Several studies have reported *O. anthropi* bacteraemia as an important cause of catheter-related bloodstream infection, particularly in immunocompromised individuals and patients with prolonged indwelling catheter use. Yu et al. analysed 15 cases of *O. anthropi* bacteraemia, portraying the organism's opportunistic nature[4]. Cieslak et al. identified *O. anthropi* as the cause of catheter-associated bacteraemia in a three-year-old child with retinoblastoma[5].

Arora et al. reported a case of *O. anthropi* septicaemia in an elderly cardiac-care-unit patient, with contaminated hospital water implicated as the likely source[6]. Braun et al. described *O. anthropi*-induced endophthalmitis after cataract surgery, necessitating intraocular lens removal[7].

These studies emphasize *O. anthropi*'s ability to cause serious infections in vulnerable patients and highlight the need for strict aseptic practices. Our isolate demonstrated resistance to multiple antibiotics — including colistin, amoxicillin-clavulanate, piperacillin-tazobactam, and cephalosporins — while remaining susceptible to imipenem, meropenem, and trimethoprim-sulfamethoxazole.

In our case, the central venous catheter likely served as the portal of entry for this opportunistic pathogen, consistent with previously reported cases where catheter colonization played a central role in infection development. Despite its relatively low virulence, *O. anthropi* demonstrates significant antimicrobial resistance; it is frequently resistant to penicillin, ampicillin, beta-lactam/beta-lactamase inhibitor combinations, chloramphenicol, tetracycline, and colistin, but often remains susceptible to fluoroquinolones, aminoglycosides, trimethoprim-sulfamethoxazole, and carbapenems[8].

This case reinforces the importance of source control: catheter removal was ultimately essential for definitive resolution, in addition to appropriate antibiotic therapy. Effective management of *O. anthropi* infections requires prompt identification, often aided by automated culture systems, given the organism's frequent misidentification as *Pseudomonas* or *Brucella* species[2].

This case emphasizes that even low-virulence organisms such as *O. anthropi* can cause persistent, clinically significant bloodstream infections, particularly in hospitalized and immunocompromised patients with indwelling devices. Early identification, appropriate antimicrobial selection, and timely removal of infected catheters remain the cornerstone of successful management.

IV. Conclusion

This case highlights that *Ochrobactrum anthropi* is an emerging nosocomial pathogen that deserves significant medical attention because of its wide environmental distribution and clinical relevance, particularly in immunocompromised patients and those with indwelling catheters or prolonged ICU stay. Its ability to survive in hospital environments and contaminate medical devices increases the risk of healthcare-associated infections. This case also emphasizes the importance of strict infection control practices, prompt culture collection before initiating antibiotics, antimicrobial stewardship, and susceptibility-guided targeted therapy in achieving successful outcomes with unusual, multidrug-resistant nosocomial pathogens.

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ORIGINAL ARTICLE

Geospatial Network Analysis of Maternal Healthcare Access and Travel Impedance in South 24 Parganas District, West Bengal

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Abstract

Background: Geographic disparities in access to emergency obstetric care contribute heavily to maternal mortality. Spatial barriers often delay life-saving clinical interventions. **Aims & Objectives:** This study aimed to model the travel network impedance from localized population centers to tertiary and secondary referral health facilities in South 24 Parganas, West Bengal. **Methodology:** A secondary spatial data analysis was conducted for the South 24 Parganas district. The geographical coordinates bounded the region between 21° 29′ N to 22° 33′ N latitude and 88° 3′ E to 89° 5′ E longitude. Eight population nodes and five institutional health hubs were mapped. QGIS software v 3.36 was used to calculate travel time matrices using existing road and waterway network vector layers. Bed counts and local population densities served as core attribute variables. **Results:** The calculated travel costs from population nodes to the closest facilities varied significantly across the geographic landscape. The minimum travel impedance recorded was 19.55 minutes. The maximum recorded travel impedance reached 125.15 minutes. Remote deltaic sectors demonstrated extreme transit times to higher-order health facilities. **Conclusion:** Severe spatial constraints restrict healthcare access for maternal populations within remote hubs of South 24 Parganas. Targeted infrastructure investments and strategic placement of emergency obstetric care facilities are crucial to bridge these geographic gaps.

Key Words: Geographic Information Systems, Health Services Accessibility, Maternal Health Services, Maternal Mortality, Travel Time

Key Message: Geospatial network modeling reveals critical gaps and extreme travel-time variation — up to 125 minutes — for maternal populations accessing secondary and tertiary referral health hubs in South 24 Parganas.

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I. Introduction

Maternal mortality remains a significant global public health concern.[1] Type II delay in reaching an appropriate medical facility is a major driver of preventable maternal deaths.[2,3] Spatial distance and poor transport infrastructure aggravate these barriers.[4] This problem is particularly acute in developing countries and rural sub-regions.[5]

The district of South 24 Parganas, with an approximate maternal mortality ratio of 110 per one lakh live births, has unique and complex geographical terrains.[6] The deltaic region contains massive networks of rivers, creeks, and waterways.[7] This estuarine terrain hinders the continuous development of standard road infrastructure. Consequently, pregnant women residing in these remote pockets face immense physical challenges during medical emergencies. They must navigate fragmented roads and seasonal river transport systems to reach healthcare facilities.[8]

Traditional administrative planning often overlooks real-world transit layouts.[9] Straight-line distance measurements do not represent true travel impedance over complex terrains.[10] Geographic Information Systems (GIS) provide advanced tools to address this limitation.[8] Network analysis techniques calculate precise travel times along actual transport vectors.[10] These models incorporate physical barriers, road classes, and transport speeds.[9] Thus, spatial modeling offers a realistic assessment of maternal healthcare accessibility.

A detailed evaluation of the existing referral network is highly necessary.[8] The evidence might allow health administrators to optimize resource distribution.[5] It also helps in placing emergency obstetric care facilities strategically.[3] Therefore, this study was conceptualized to fill the existing gaps in regional transport planning for maternal health. The study aimed to analyze the spatial network accessibility and determine travel time impedance from various population centers to designated secondary and tertiary healthcare hubs in the South 24 Parganas district.

II. Materials and Methods

Study Design and Study Area

This study was a secondary data analysis using geospatial modeling techniques. It was conducted for the South 24 Parganas district located in West Bengal, India. The geographic boundaries of the study area extend from 21° 29′ N to 22° 33′ N latitude and 88° 3′ E to 89° 5′ E longitude.[6] This district includes a mix of peri-urban plains and the unique riverine tracts of the Sundarbans delta.[7] (Figure 1)

Network Nodes and Hubs

The spatial network model incorporated specific geographic points to assess travel friction.[9] Eight distinct population clusters were selected and designated as origin nodes. These nodes represented major community settlements across the district geography. The selected nodes included Budge Budge, Sonarpur, Jaynagar, Mograhat, Mathurapur, Kulpi, Falta, and Basanti. Five primary institutional health facilities were selected and designated as destination hubs. These hubs included Diamond Harbor Medical College, Baruipur Sub-Divisional Hospital (SDH), Canning Sub-Divisional Hospital (SDH), Sarisha Block Primary Health Centre (BPHC), and Padmahat Block Primary Health Centre (BPHC). (Figure 1)

Study Variables and Data Sources

Multiple spatial and non-spatial variables were integrated into the analysis database. The primary non-spatial attribute was total bed counts at each health facility hub.[6] Local population numbers within each settled locality node were included as a demand variable. The primary spatial data layers consisted of road networks, navigable waterways, and point locations of health hubs. The secondary spatial layers, transport networks, and demographic variables were sourced from official public domain repositories and institutional health records.

Data Analysis and Spatial Modeling

Spatial network analysis was executed utilizing open-source QGIS software version 3.36. It was used to compile the geographic database and run network simulations. A digital network graph was constructed using the vector

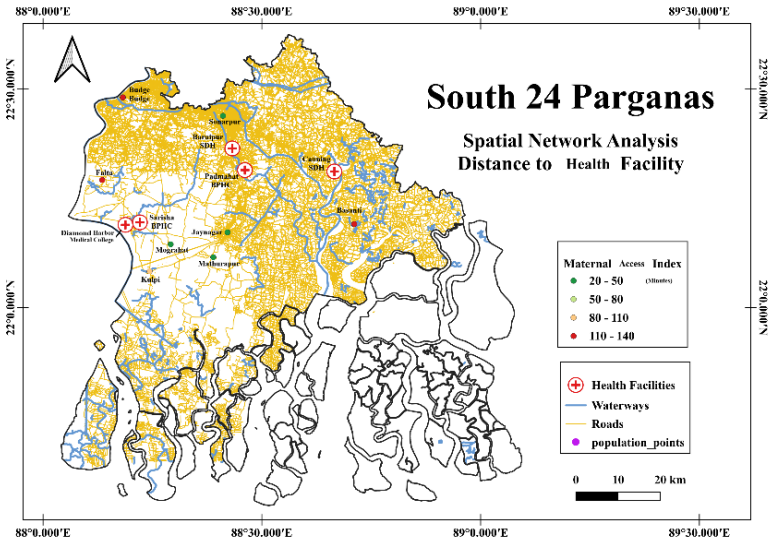


Figure 1: *Spatial Network Analysis and Travel Distance to Health Facilities in South 24 Parganas.*

road networks and waterway layers. Travel speeds were assigned to different road categories based on standard transport guidelines. Waterway transit parameters were added to represent ferry crossings in deltaic zones.

The Network Analysis toolbox was utilized to run an origin-destination cost matrix.[9] The algorithm calculated the shortest travel paths along the network vectors.[10] The accumulated travel time in minutes was computed from each of the eight origin nodes to each of the five destination hubs. The mathematical costs accounted for land-water transitions and topological connectivity. The resulting matrix values were mapped to produce a Maternal Access Index layout. The findings were visually represented using a graded colour scheme to indicate variations in transit times across the district.

The accumulated travel time in minutes was computed from each of the eight origin nodes to each of the five destination hubs to establish a comprehensive multi-destination impedance matrix. Subsequently, a secondary proximity filter was applied to isolate and map only the single least time-consuming path from each origin node to its nearest accessible healthcare facility.

Ethical Considerations

This research was based entirely on the analysis of secondary data layers. The dataset did not contain any individual human participant identifiers. No clinical records or personal health information were accessed during the data compilation. Therefore, this study did not require any formal ethical clearance or institutional review board permission.

III. Results

The spatial distribution of health facilities and the travel time categories across South 24 Parganas were mapped systematically. The geographic layout of transport vectors, waterways, and facility catchments is shown in Figure 1.

The network analysis quantified the travel impedance values between all nodes and hubs. The complete matrix of travel times in minutes is presented in Table 1.

The Maternal Access Index is divided into four main travel time tiers. The green points show the shortest travel times between 20 and 50 minutes. The light green points indicate moderate travel times between 50 and 80 minutes. The orange points highlight high travel times between 80 and 110 minutes. The red points mark extreme travel times ranging from 110 to 140 minutes. (Figure 1)

Table 1: *Travel Time Impedance Matrix from Population Nodes to Health Hubs (Minutes)*

Origin ID	Population Node	Hub 1: Diamond Harbor Medical College	Hub 2: Baruipur SDH	Hub 3: Canning SDH	Hub 4: Sarisha BPHC	Hub 5: Padmahat BPHC
1	Budge Budge	33.83	76.17	125.15	40.75	79.87
2	Sonarpur	25.82	79.17	99.30	19.55	70.77
3	Jaynagar	30.67	93.64	109.15	28.46	80.65
4	Mograhat	75.87	70.16	119.46	82.79	83.29
5	Mathurapur	89.82	19.95	64.32	89.58	33.07
6	Kulpi	54.04	50.57	67.65	47.20	37.58
7	Falta	53.72	64.75	80.27	46.88	51.76
8	Basanti	120.84	77.61	35.19	114.00	69.74

The lowest travel time recorded in the entire network is 19.55 minutes. This minimum time occurs from the origin node 2 to the destination hub 4. The highest travel time recorded in the network is 125.15 minutes. This maximum impedance occurs from origin node 1 to destination hub 3. Origin node 8 shows a low travel time of 35.19 minutes to destination hub 3. However, origin node 8 shows an extreme travel time of 120.84 minutes to destination hub 1. Origin node 4 maintains a travel time above 70 minutes across all five available destination hubs. (Table 1)

The time-optimized spatial network pathway analysis mapped the actual physical transit pathways along the road and waterway infrastructure of the South 24 Parganas district. The multi-destination geographic distribution model highlights a stark regional imbalance. (Figure 2) Figure 2a describes the exhaustive network of all potential spatial paths. The time-optimized pathways in Figure 2b demonstrate that rapid healthcare access remains localized. The most efficient transit routes (under 29 minutes) are heavily restricted to the western and northwestern perimeters, leaving communities adjacent to the southern deltaic networks isolated, facing significantly higher travel times to reach secondary or tertiary clinical infrastructure.

IV. Discussion

The main objective of this study was to analyze spatial network accessibility and determine travel time impedance from population nodes to designated health hubs in South 24 Parganas. The findings showed a wide variation in travel impedance across the district.

Several factors explain the significant differences in travel times. The estuarine geography of South 24 Parganas creates natural obstacles for land transport.[7] Localities like Basanti are separated from major urban centers by rivers and creeks. Relying on ferry connections increases transit times significantly.[6] This reliance explains the high travel cost of 120.84 minutes from Basanti to Diamond Harbor Medical College. Conversely, sectors with contiguous highway networks show much lower travel impedance. For example, Sonarpur shows a short travel time of 19.55 minutes to Sarisha BPHC due to better road connectivity. Fragmented infrastructure directly compromises maternal health outcomes.[5] Long transit times prolong the second delay during obstetric emergencies.[2] This delay increases the risk of birth complications and maternal mortality.[4] Pockets facing travel times above 80 minutes require urgent attention. These high-impedance zones leave pregnant women vulnerable when complications arise.[8]

Subsequently, when the network is optimized to model transit strictly to the closest available health facility (Figure 2b), a different

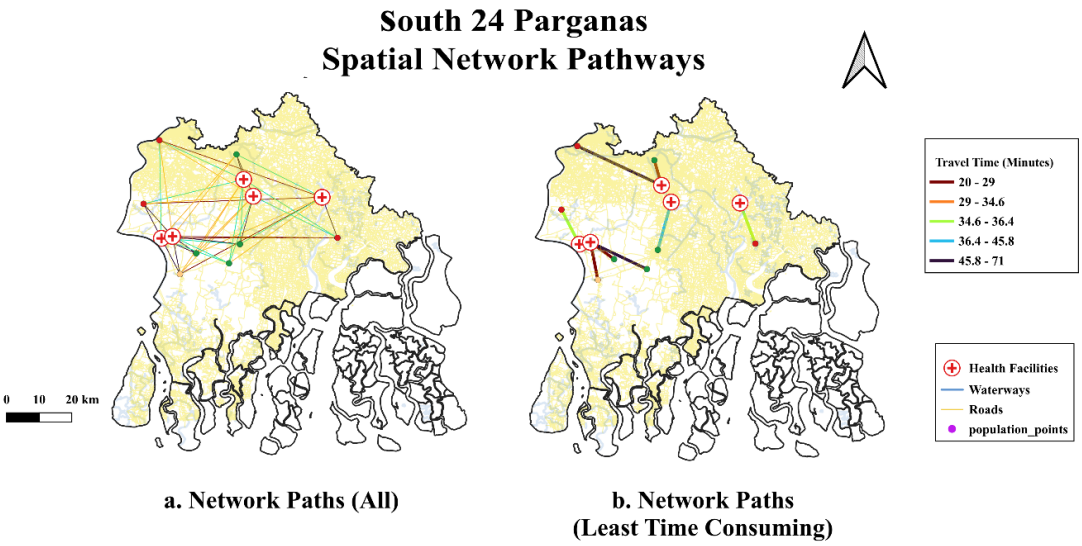


Figure 2: *Time-Optimized Spatial Network Path Analysis and Geographic Accessibility Tiers across South 24 Parganas, West Bengal.*

layer of spatial inequality emerges. Even under this ideal-case scenario, the safety net is not uniform; while western border populations enjoy highly efficient access within 29 minutes, other northern and sub-deltaic communities still experience localized infrastructure friction and bottlenecks that stretch nearest-facility transit up to 71 minutes. This implies that proximity on a map does not automatically equate to rapid clinical access when the underlying road vectors are fragmented.

This study used actual road and waterway network paths instead of simple straight-line distances. This approach provided a realistic model of travel impedance over difficult deltaic terrain. The dual-model approach of integrating a comprehensive multi-destination matrix alongside a proximity-filtered nearest-hub analysis significantly enhances the accuracy and utility of the geographic model for real-world referral planning.

Thus, localities with travel times exceeding 100 minutes need immediate decentralized care options. Specialized obstetric stabilization units should be set up at riverine transit points. Multi-modal transport networks must be improved. Upgrading ferry services and building bridges will reduce travel friction significantly. Lastly, future research should incorporate seasonal variables into network models. Analyzing transit variations during heavy monsoons will provide deeper insights. Researchers should also look at how facility capacities affect patient choices. Combining spatial travel costs with clinical quality data will help optimize maternal health planning.

V. Conclusion

This study highlights major geographic disparities in maternal healthcare access across South 24 Parganas. Network modeling revealed that travel times to referral hubs range from under 20 minutes to over two hours. The riverine landscape creates significant transit barriers for remote communities. These findings emphasize that straight-line distance models underestimate actual travel barriers. Reducing these spatial inequities requires targeted infrastructure investments and decentralized emergency obstetric facilities. Implementing these spatial adjustments is essential to eliminating transit delays and lowering maternal mortality.

Limitations of the Study

The network analysis assumed static travel

speeds along transport vectors. It did not account for real-time traffic congestion or seasonal weather disruptions. Monsoonal flooding often slows transit speeds or disrupts ferry services completely in the Sundarbans. Furthermore, the model did not include the use of private transport modes or emergency ambulance availability.

Authors' Contribution

All authors contributed substantially to this study, from conceptualization, study design, literature search, data curation and data analysis to manuscript writing and review. All of them are guarantors of this work.

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The authors did not receive any support from any source.

Conflict of Interest

The authors have no conflicts of interest regarding this work.

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Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

The authors did not use AI at any stage of this study.

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Original Article

Assessment of Stability of Fluoride Vials Over a Time Period

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Abstract

Background: Blood glucose estimation is one of the commonest and most widespread tests in any clinical laboratory. Rising concern about diabetes mellitus and metabolic syndrome emphasizes correct evaluation of blood glucose. Grey-top fluoride vials are used in most laboratories, though their stability over time is not well characterized. **Objectives:** To predict the stability of grey-top sodium fluoride (NaF) vials, so as to minimize erroneous glucose results when laboratory turnaround time is increased or repeat measurement from the same vial is required. **Methods:** Twelve retained blood glucose samples (across four clinically relevant glucose levels) were rerun at six time points — baseline, 24, 48, and 72 hours, and 7 and 14 days after collection — on a Vitros 4600 dry chemistry system using the glucose oxidase-peroxidase method. **Results:** A statistically significant difference from baseline was observed at 24 hours (paired t-test, p=0.005), whereas differences at 48 hours, 72 hours, 7 days, and 14 days were not statistically significant ($p>0.05$). Mean $\pm$ SD glucose values were 155.50 $\pm$ 65.13 mg/dL (24 h), 153.33 $\pm$ 63.49 mg/dL (48 h), 153.42 $\pm$ 63.65 mg/dL (72 h), 153.42 $\pm$ 63.43 mg/dL (7 days), and 154.25 $\pm$ 63.60 mg/dL (14 days), against a baseline of 153.75 $\pm$ 64.25 mg/dL. **Conclusion:** A statistically significant difference from baseline was detected at 24 hours, but the absolute mean difference was small (1.75 mg/dL). No statistically significant differences were detected at 48 hours, 72 hours, 7 days, or 14 days. These findings suggest that, under the storage conditions used in this study, the observed statistical change at 24 hours should be interpreted cautiously and should not by itself establish a universal 24-hour stability limit for all grey-top fluoride specimens.

Key Words: Glucose, Grey-top vials, Sodium fluoride, Stability

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Ethics Committee Clearance: Not taken — the study used retained, de-identified blood glucose samples with no patient names, ages, or genders recorded, and sample handling followed Biomedical Waste Management guidelines (as stated in the manuscript)

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Conflict of Interest: None declared (as stated in the manuscript)

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I. Introduction

Diabetes mellitus remains a rising health concern in developing and developed countries alike. It is a metabolic disease characterized by increased blood glucose levels with absolute or relative insulin deficiency or insulin resistance, and related complications. Community studies indicate that the growing diabetic population is creating an increasing burden of ill health on society[1]. With significant lifestyle changes, healthcare professionals remain concerned about the accuracy and precision of blood glucose measurement — not only for known diabetics, but also for early diagnosis of pre-diabetics, which may lead to better clinical outcomes.

Assurance of the integrity and reliability of clinical laboratory results is a necessity. Challenges to quality results include preanalytical, analytical, and postanalytical errors, as well as biological variation. According to National Academy of Clinical Biochemistry guidelines, in vitro glycolysis must be prevented to avoid erroneous blood glucose estimation[2]. Blood glucose is usually measured in vials containing sodium fluoride (NaF) and sodium oxalate: fluoride, a competitive inhibitor of enolase, inhibits glycolysis and preserves the true glucose value, while oxalate acts as an anticoagulant. The action of NaF is slow, taking almost 4 hours from collection to show its antiglycolytic effect, and remains stable thereafter for 2–3 days[3,4]. A study of 200 pregnant women assessed for gestational diabetes mellitus concluded that fluoride-citrate vials are a better alternative for diagnosis, investigating fluoride, citrate, and ice as antiglycolytic agents[5] — citrate being an inhibitor of phosphofructokinase in the glycolytic pathway.

Vial stability is a prerequisite in a clinical laboratory setting, both for cross-checking values to rule out preanalytical, analytical, and postanalytical error or normal biological variation, and for accommodating delays in sample processing that may occur in practical hospital settings without altering the result and consequent diagnosis. Since glucose is one of the commonest laboratory parameters, any measurement discrepancy may lead to a wrong diagnosis and a serious downstream outcome. Fluoride-treated specimens are more effective than heparin-treated specimens, as the rate of decrease in glucose concentration after the first 4 hours is much slower in fluoride vials[4]. This study aims to predict the stability of grey-top NaF vials, in order to remove erroneous results in blood glucose estimation even where laboratory turnaround time increases.

Objective

To predict the stability of grey-top NaF vials, so as to remove erroneous results in blood glucose estimation even under conditions of increased turnaround time, or where a repeat measurement from the same vial is required due to machine-related technical error.

II. Materials and Methods

The study was conducted in the Biochemistry Laboratory of JMN Medical College, Chakdaha, as a trial run on retained samples, to predict the stability of grey-top glucose vials over a time period. Four levels of blood glucose value of clinical significance were selected — 90, 120, 150, and 250 mg/dL — with a total of 12 samples taken such that three samples in each group had similar glucose levels. Sampling was by convenience, and samples were selected to include fasting, post-prandial, and random blood glucose specimens. Approximately 2 mL of blood was present in all BD grey-top vials, containing potassium oxalate and sodium fluoride. Haemolysed and clotted samples were excluded.

Samples were rerun at 24 hours, 48 hours, 72 hours, 7 days, and 14 days after collection. Glucose was estimated on a Vitros 4600 Dry Chemistry system using the glucose oxidase-peroxidase (GOD-POD) method. Samples were stored at 2–8°C in the Biochemistry Clinical Laboratory freezer between runs.

III. Results

Glucose values in grey-top vials were assessed at six time points (n=12); only the 24-hour measurement differed significantly from baseline. Mean $\pm$ SD glucose levels were 155.50 $\pm$ 65.13

mg/dL (24 h), 153.33 ± 63.49 mg/dL (48 h), 153.42 ± 63.65 mg/dL (72 h), 153.42 ± 63.43 mg/dL (7 days), and 154.25 ± 63.60 mg/dL (14 days), against a baseline of 153.75 ± 64.25 mg/dL; coefficients of variation were 41.89%, 41.41%, 41.49%, 41.34%, and 41.23% respectively. The mean difference from baseline was 1.75 mg/dL at 24 h (p=0.005), 0.42 mg/dL at 48 h (p=0.480), 0.33 mg/dL at 72 h (p=0.368), 0.33 mg/dL at 7 days (p=0.457), and 0.5 mg/dL at 14 days (p=0.293).

Table 1: *Glucose measurements over time in retained grey-top fluoride vials (n=12)*

Time point	Mean ± SD (mg/dL)	Difference; *p* value
Baseline	153.75 ± 64.25	—
24 h	155.50 ± 65.13	1.75; 0.005
48 h	153.33 ± 63.49	-0.42; 0.480
72 h	153.42 ± 63.65	-0.33; 0.368
7 days	153.42 ± 63.43	-0.33; 0.457
14 days	154.25 ± 63.60	0.50; 0.293

IV. Discussion

This study aimed to estimate the stability of grey-top glucose vials over six time points. Only the 24-hour measurement differed significantly from baseline (p=0.005); the differences at 48 hours, 72 hours, 7 days, and 14 days were not statistically significant. The mean glucose level increased by 1.75 mg/dL at 24 hours, decreased by 0.42 mg/dL at 48 hours and by 0.33 mg/dL at 72 hours and 7 days, and increased by 0.50 mg/dL at 14 days. The observed fluctuations were small and may reflect analytical or preanalytical variation; the study design does not allow the specific cause to be established. Notably, the difference remained similar from the third to the seventh day, suggesting that glucose measurements remained relatively stable across those later time points under the study storage conditions.

The statistically significant 24-hour difference was small (1.75 mg/dL) and is unlikely to be clinically important for glucose interpretation in routine laboratory medicine; such minor variation would not typically change a diagnosis or clinical decision. The different glucose-level groups showed broadly similar patterns of change over time. This finding is consistent with Imtiaz et al. (2019), who reported a mean difference in glucose level of 1.2 mg/dL after 24 hours[6].

Glucose estimation from grey-top vials is therefore preferable within a day of collection, though delayed estimation does not appear to carry major clinical significance.

Limitation

The main limitation of this study is its small sample size. A study with a larger sample size is required to give better insight and provide further clarity for laboratory medicine and improved patient care.

V. Conclusion

Under the storage conditions used in this study, the observed mean glucose values remained relatively stable across the study period, with a statistically significant but small difference at 24 hours. Larger studies are needed before defining a universal stability limit for grey-top fluoride specimens.

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Case Study / Case Report

Anomalous Malrotation of Intestine with Ileocecal Junction in the Left Iliac Fossa: A Cadaveric Case Report

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Abstract
Intestinal malrotation is a rare congenital anomaly resulting from abnormal rotation and fixation of the midgut during embryological development. While most cases are identified clinically in neonates or children, incidental findings during cadaveric dissection provide valuable insights into anatomical variations. We report a unique case of anomalous malrotation in which the ileocecal junction was located in the left iliac fossa, discovered during undergraduate teaching dissection at JMN Medical College.

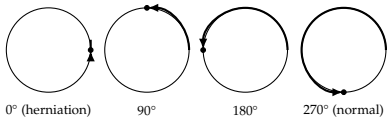
Key Words: Ileocecal junction, Intestinal malrotation, Left iliac fossa, Midgut rotation

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Ethics Committee Clearance: Not reported
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Conflict of Interest: None declared
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I. Introduction

Normal embryological development of the midgut involves a 270° counterclockwise rotation around the superior mesenteric artery, resulting in the caecum and ileocecal junction being positioned in the right iliac fossa[1]. Malrotation occurs when this process is incomplete or abnormal. Such anomalies are rarely documented in cadaveric studies, making this case noteworthy for both anatomical education and clinical relevance.

Figure 1: Stages of Normal Midgut Rotation



Schematic of the four canonical stages of midgut rotation around the superior mesenteric artery axis, from pre-rotation herniation (0°) through to the normal post-rotation position (270°, caecum in right iliac fossa). Non-rotation and incomplete rotation, as in the present case, arrest at an intermediate stage.

II. Case Description

During routine undergraduate cadaveric dissection of an adult male cadaver in the Department of Anatomy at JMN Medical College, the caecum and ileocecal junction were found in the left iliac fossa instead of the right. The ascending colon was shortened and displaced, with small intestinal loops occupying atypical positions. No pathological adhesions or surgical scars were noted, supporting a congenital anatomical variant without evidence of previous abdominal surgery. The finding was demonstrated to undergraduate students, highlighting the importance of recognizing anatomical variations that may have clinical implications.

The small intestine was located on the right side of the abdomen, while the large intestine was situated on the left side. The superior mesenteric vein was found to the left of the superior mesenteric artery. No abnormal peritoneal band (Ladd band) was identified. **Thoracic and cardiac disposition were not assessed as part of this dissection**; the abdominal findings alone cannot establish whether this cadaver had accompanying situs inversus or other laterality anomalies.

III. Discussion

Malrotation is typically diagnosed in living patients due to symptoms such as volvulus or obstruction[2]. In cadaveric studies, such anomalies provide direct evidence of developmental variation and enrich anatomical teaching. The presence of the ileocecal junction in the left iliac fossa is extremely rare and may complicate clinical diagnosis in living patients, as it can mimic left-sided abdominal pathologies[3]. Documentation of such cases enhances awareness among clinicians and anatomists, ensuring better interpretation of imaging and surgical findings.

Anomalous malrotation of the intestine with the ileocecal junction in the left iliac fossa indicates a congenital anatomical defect in which the midgut fails to rotate and fixate properly during foetal development. In this specific variant — often categorized under intestinal non-rotation — the usual 270° counterclockwise rotation is incomplete, typically arresting at 90°.

Torres and Ziegler[4] described the embryological basis and clinical spectrum of malrotation, emphasizing that abnormal positioning of the caecum and ileocecal junction can lead to diagnostic confusion and potentially life-threatening complications. In our cadaveric case, the ileocecal junction was found in the left iliac fossa, an exceptionally rare anatomical variation that underscores the diversity of malrotation presentations.

Adult presentations of malrotation are uncommon but increasingly recognized. Matzke et al.[5] reported successful laparoscopic management of adult malrotation with cocoon deformity, highlighting that surgical intervention remains the definitive treatment when symptomatic. Similarly, von Flüe et al.[6] documented both acute and chronic presentations of non-rotation in adults, reinforcing that malrotation may remain silent until adulthood or present with nonspecific abdominal complaints. Wang and Welch[7] earlier emphasized that anomalies of intestinal rotation in adolescents and adults often mimic other abdominal pathologies, complicating diagnosis.

The clinical relevance of our cadaveric finding lies in its potential to mimic left-sided abdominal conditions. Jacobs[8] and Ozick et al.[9] discussed diverticulitis and diverticular disease as common causes of left iliac fossa pain. In a living patient, a left-sided ileocecal junction could easily be misinterpreted as sigmoid or descending colon pathology, leading to misdiagnosis and inappropriate management. Thus, awareness of such rare anatomical variations is crucial for surgeons, radiologists, and clinicians.

From an educational perspective, cadaveric dissection remains invaluable in exposing medical students to anatomical variations beyond textbook norms. Documenting such anomalies enriches anatomical literature and bridges embryological theory with clinical practice.

Limitations

This report describes abdominal visceral disposition only. The thoracic cavity and cardiac

position were not assessed during the present dissection; therefore, a co-occurring laterality anomaly such as situs inversus or heterotaxy cannot be confirmed or excluded. This limitation should be considered when interpreting the anatomical findings.

IV. Conclusion

This cadaveric case highlights a rare variant of intestinal malrotation with the ileocecal junction in the left iliac fossa. Such findings underscore the importance of cadaveric dissection in medical education, offering students first-hand exposure to anatomical variations that may have significant clinical relevance.

Learning Points

- Rare anatomical variation: the ileocecal junction in the left iliac fossa due to malrotation is an uncommon finding.
- Educational value: cadaveric dissection provides invaluable opportunities to observe and document rare anomalies.
- Clinical relevance: such variations may mimic left-sided abdominal pathologies, complicating diagnosis in living patients.
- Embryological insight: reinforces the importance of understanding midgut rotation and its deviations.
- Documentation importance: recording anomalies contributes to medical literature and enhances anatomical and clinical knowledge.

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Original Article

A Study on Consumer Awareness and Disposal Practice of E-waste and Their Determinants in Kolkata, West Bengal

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Abstract

Background: The growing burden of e-waste is an emerging public health concern. Consumers are important stakeholders, and percolating correct knowledge and generating awareness of e-waste are critical for proper e-waste handling and management, to minimize hazards and guide e-waste into a circular economy. This study aimed to assess consumer awareness and disposal practices at the individual level and their determinants. **Methods:** This online survey was conducted through a Google Form comprising a pre-designed, pre-tested structured questionnaire on e-waste-related awareness and disposal practices among adult residents of Kolkata. A total of 439 respondents submitted their responses from September to October 2020. Multivariable logistic regression was used to identify factors associated with satisfactory awareness levels and correct disposal practices. **Results:** A satisfactory level of awareness was found in 79% of participants. Younger age ($\leq$35 years vs. $>$35 years, AOR: 3.62), higher educational qualification (graduate and/or above vs. higher secondary or below, AOR: 3.73), and professional employment (vs. homemakers, AOR: 2.66) were associated with significantly higher awareness. Selling or exchanging devices during new purchases was the predominant disposal method for old electronics; only 4.1% deposited e-waste at designated e-waste collection points. **Conclusion:** Despite a satisfactory level of awareness, very few respondents disposed of old and unused e-products at e-collection centres. Generating awareness through mass media, introducing attractive take-back schemes, engaging municipal bodies for door-to-door collection, and building more e-collection centres are important measures for proper e-waste management.

Key Words: Consumer awareness, Disposal, Electronic waste, E-waste, WEEE

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Source(s) of Support: Not reported
Conflict of Interest: None declared

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I. Introduction

The growing burden of electrical and electronic waste (e-waste) is an emerging concern of this century[1,2]. E-waste is defined as items of electrical and electronic equipment (EEE) and their parts that have been discarded by the owner without intent of re-use, regardless of whether the item is still functional[1]. The Global E-waste Monitor 2020 reported 53.6 million tonnes of e-waste generated globally in 2019, of which Asia contributed 24.8 million tonnes; the global burden is projected to reach 74.7 million tonnes by 2030, yet only 17.4% was formally collected and recycled in 2019[3]. E-waste is a complex mixture of hazardous materials (heavy metals, brominated flame retardants, polyaromatic hydrocarbons) and recoverable precious elements[2,4-8]. In low- and middle-income countries, e-waste is often recycled crudely in household settings, exposing the general population — particularly women and children — to hazardous effects[7-11].

India is the third-largest e-waste generator globally, with only about a quarter of its waste stream handled through formal recycling capacity[3], and roughly 16% of national e-waste is generated by individual households[16]. Lack of awareness, weak enforcement of e-waste law and policy, and the absence of an organized collection system are identified as key barriers to proper e-waste management in India[4,7].

Kolkata, among the top ten contributors to India's e-waste stream and a major IT and business hub of the eastern zone with a population of over 14 million[10], is expected to see this burden grow further in the post-pandemic period. Against this backdrop, the current study aimed to assess consumer awareness and disposal practice of e-waste among residents of Kolkata, West Bengal.

II. Materials and Methods

This was an online, cross-sectional survey conducted via a pre-designed, pre-tested, structured Google Form questionnaire covering e-waste-related awareness and disposal practices. Adult residents of Kolkata were the target population. A total of 439 adult residents of Kolkata submitted valid responses between September and October 2020. Multivariable logistic regression was used to identify factors independently associated with (a) a satisfactory awareness level and (b) correct disposal practice.

III. Results

Of 439 respondents, the majority correctly associated multiple categories of items with e-waste: 61.7% identified irreparably damaged gadgets, functioning-but-unwanted devices, and broken electronic parts collectively as e-waste. Awareness that e-waste is hazardous to both humans and the environment was reported by 56.5%, while 5.7% did not know whether e-waste was hazardous at all. Most respondents (73.1%) believed e-waste needs special treatment before disposal, and 59.5% correctly identified that e-waste contains precious recoverable elements. Awareness of formal infrastructure and policy was markedly lower: only 12.5% knew of an e-waste collection centre in their city, and just 6.8% were aware of any e-waste-related law in India. **Table 1**

Fifteen participants (3.4%) scored the maximum on the awareness scale, and 347 (79%) scored $\geq$5, meeting the threshold for satisfactory awareness. In multivariable logistic regression, younger individuals ($\leq$35 years vs. >35 years, AOR: 3.62; 95% CI: 2.05–6.39), higher educational qualification (graduate and/or above vs. higher secondary or below, AOR: 3.73; 95% CI: 2.05–6.78), and professional jobholders (vs. homemakers, AOR: 2.66; 95% CI: 1.05–6.75) were found to have significantly higher awareness. **Table 2** The non-significant Hosmer-Lemeshow test (0.312) indicated good model fit, with 9.4–14.6% of the variation in satisfactory awareness explained by the model (Cox & Snell R^2 = 0.094, Nagelkerke R^2 = 0.146).

About half of the respondents usually sell or exchange old or unused devices when buying new products, while 20% store unused gadgets at home; only 18 participants (4.1%) deposited e-waste at e-waste collection points. None of the sociodemographic factors examined were associated with correct disposal practice in this study ($p>0.05$). **Table 3** A large

proportion (322, 73%) considered that financial responsibility for e-waste recycling and disposal should be shared among government, producers, traders, and citizens. Most respondents (356, 81.1%) reported rarely or never encountering e-waste-related advisories in print, digital, or social media, and salespersons never shared such information with 376 (85.6%) of respondents at the point of purchase. Only about a quarter of participants considered their own e-waste knowledge satisfactory, and about 90% (n=394) expressed a need for further education on the subject.

IV. Discussion

The current study showed that 347 (79.0%) of 439 participants had a satisfactory awareness level (total score $\geq$5). This is considerably higher than previously reported figures from other parts of India: Sivathanu et al.[4] reported 58.5% awareness among consumers in Pune; a 2013 Lucknow study among young adults reported 60.8% and 37.5% with medium and high knowledge respectively[11]; and a facility-based study among undergraduate medical students in Delhi found 77.3% aware, with 42.7% having adequate knowledge[17]. Miner et al.[9] reported 67.5% awareness in a 2018 Nigerian household survey.

In the present study, 61.7% considered broken, damaged, or simply unused electronic products collectively as e-waste. Around 4% considered e-waste non-hazardous and 5.7% did not know of any hazardous effect, while 19.6% considered it hazardous without knowing the specific effect. Shah et al.[7] in Ahmedabad similarly found 28% did not perceive e-waste as hazardous, and 37% considered it hazardous without being able to specify the effects; Miner et al.[9] found 68% aware of toxic effects, with 60.5% and 68% perceiving harm to health and environment respectively. About 73.3% of our participants felt e-waste needs special treatment before disposal, higher than the 50.9% reported by Miner et al.[9]

A majority of our participants (87.5% and 93.2% respectively) were unaware of any e-waste collection centre in Kolkata or any e-waste-related law or policy in India. Shah et al.[7] similarly reported only 37% aware of formal e-collection points and 11% aware of any government policy — despite more than six years having passed since that study, and despite e-waste handling and management law having been in place in India since 2011, awareness of formal infrastructure and policy remains low even among a better-educated study population such as ours.

Younger age, higher educational qualification, and professional employment were associated with better awareness in multivariable analysis, consistent with findings reported in previous studies[6,7,11,18]. Respondents who encountered e-waste advisories in media, or who were informed by salespersons at purchase, tended to have better awareness than those who rarely or never did. Only 4% of respondents used formal e-collection centres, while 19.4% still store old gadgets at home — broadly consistent with other Indian studies reporting 13–26% home storage[6,7] and around 18% formal disposal[7]. Nearly half sell or exchange old devices during new purchases, suggesting that consumers perceive continued monetary value in old electronics and are reluctant to discard them without compensation. Participants used their last mobile phone, PC, and TV for a median of 3, 4, and 8 years respectively, with access to newer technology cited as the predominant reason for replacement[7]. A significant proportion (73%) felt financial responsibility for e-waste recycling should be shared among citizens, producers/dealers, and government equally, though a wider community-based survey across all socioeconomic strata would be needed to confirm this finding more broadly.

Limitation

Being an online survey with voluntary, non-probabilistic sampling, this study could only capture the awareness and disposal practice of people with internet access and generally better educational status, which may explain the relatively high awareness observed. To minimise non-response, the number of questions was kept to a minimum, leaving several domains unexplored.

Item	n (%)
What counts as e-waste (select all that apply)	
Irreparably damaged gadgets	114 (26.0)
Unwanted but functioning devices	13 (3.0)
Broken electronic parts	7 (1.6)
All of the above	271 (61.7)
Don't know/not sure	34 (7.7)
Perceived hazard of e-waste	
Hazardous, specific effect unknown	86 (19.6)
Hazardous to humans	27 (6.1)
Hazardous to environment	36 (8.2)
Hazardous to both	248 (56.5)
Not hazardous	17 (3.9)
Don't know/not sure	25 (5.7)
Needs special treatment before disposal	
Yes	321 (73.1)
No	13 (3.0)
Don't know/not sure	105 (23.9)
Contains precious/recoverable elements	
Yes	261 (59.5)
No	31 (7.1)
Don't know/not sure	147 (33.5)
Aware of an e-collection centre in their city	
Yes	55 (12.5)
No	384 (87.5)
Aware of e-waste law in India	
Yes	30 (6.8)
No	409 (93.2)

Table 1: *Distribution of study participants according to level of awareness regarding e-waste (n=439)*

Parameter	Satisfactory awareness, n (%)	AOR (95% CI)	p
Age			
≤35 y (n=207)	183 (88.4)	3.62 (2.05–6.39)	<0.001
>35 y (n=232)	164 (70.7)	1 (Ref)	
Education			
Higher sec. or below (n=54)	30 (55.6)	1 (Ref)	
Graduate+ (n=385)	317 (82.3)	3.73 (2.05–6.78)	0.010
Occupation			
Professional (n=134)	119 (88.8)	2.66 (1.05–6.75)	0.040
Homemaker (n=31)	20 (64.5)	1 (Ref)	

Table 2: *Factors significantly associated with satisfactory awareness (non-significant covariates omitted for space; N=439)*

V. CONCLUSION

A significant proportion of study participants had satisfactory awareness regarding e-waste. However, correct disposal practice remains low and is driven mainly by exchange schemes rather than formal e-waste collection. Very few respondents were aware of current e-waste policy or the location of collection points in Kolkata. Building consumer awareness through social, digital, and print media, and at the point of purchase; motivating household-level segregation; introducing door-to-door collection; promoting take-back policies for all e-waste categories; and expanding e-collection infrastructure are all needed to address the growing burden of e-waste.

Parameter	Correct disposal, n (%)	OR (95% CI)	p
Age			
≤35 y (n=207)	152 (73.4)	1 (Ref)	
>35 y (n=232)	174 (75.0)	1.09 (0.71–1.67)	0.486
Gender			
Male (n=292)	213 (72.9)	1 (Ref)	
Female (n=147)	113 (76.9)	1.23 (0.78–1.96)	0.375
Education			
Higher sec. or below (n=54)	38 (70.4)	1 (Ref)	
Graduate+ (n=385)	288 (74.8)	1.25 (0.67–2.34)	0.486

Table 3: *Factors associated with correct disposal practice — none reached statistical significance (full occupation-category breakdown in original dataset, N=439)*

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